

Earthquake Stabilized Dialysis System for Patient Safety During Seismic Events

Detailed Daily Work Report (June 14 - June 20)

This project focuses on developing a safety-oriented dialysis support system capable of protecting patients and medical equipment during earthquake events. The proposed solution uses seismic sensors, IoT communication, monitoring software, and emergency alert mechanisms to reduce risks and improve patient safety.

June 14, 2026

Topic: Earthquake Detection Study

Work Done: Studied earthquake detection principles and analyzed different seismic sensors such as accelerometers and vibration sensors. Evaluated their suitability for detecting ground motion inside hospitals and dialysis centers.

June 15, 2026

Topic: Stabilization Framework Design

Work Done: Designed a stabilization framework that can reduce equipment movement during seismic activity. Examined methods for protecting dialysis machines and preventing accidental disconnection of medical components.

June 16, 2026

Topic: IoT Monitoring Module Planning

Work Done: Prepared the design of an IoT-based monitoring module capable of collecting real-time sensor data and transmitting it to a central monitoring platform for continuous observation.

June 17, 2026

Topic: Patient Safety Alert System

Work Done: Designed an emergency alert mechanism that generates alarms, notifications, and warnings for healthcare staff when seismic activity exceeds predefined safety thresholds.

June 18, 2026

Topic: Emergency Response Workflow

Work Done: Developed a workflow describing how the system responds during an earthquake, including machine protection, patient monitoring, emergency notifications, and safe operational procedures.

June 19, 2026

Topic: System Integration Planning

Work Done: Planned the integration of seismic sensors, microcontroller units, communication modules, databases, and monitoring software to ensure smooth data exchange and coordinated operation.

June 20, 2026

Topic: Review and Testing Strategy

Work Done: Reviewed the complete system architecture, updated project documentation, identified testing scenarios, and prepared validation strategies for future implementation and evaluation.